Introduction to Neuroscience: Behavioral Neuroscience

Neuroethology, Comparative Neuroscience, Natural Neuroscience

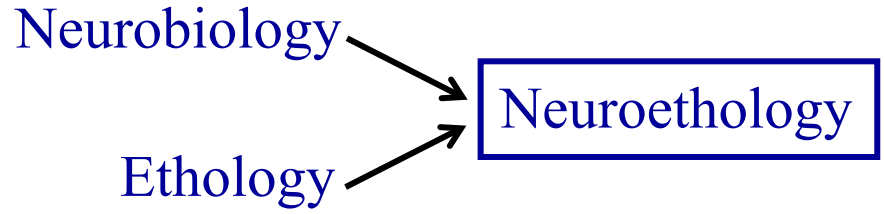
Nachum Ulanovsky

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2025-2026, 2nd semester

Principles of Neuroethology

Neuroethology seeks to understand the mechanisms by which the central nervous system controls the natural behavior of animals.



- **Focus on Natural behaviors:** Choosing to study a well-defined and reproducible yet natural behavior (either Innate or Learned behavior)
- **Need to study thoroughly the animal's behavior, including in the field:** Neuroethology starts with a good understanding of Ethology.
- **If you study the animals in the lab, you need to keep them in conditions as natural as possible**, to avoid the occurrence of unnatural behaviors & disappearance of natural behaviors.
- **Krogh's principle**

Krogh's principle



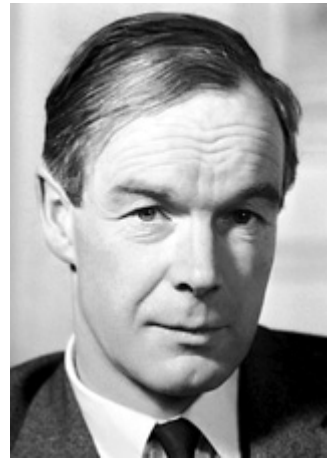
August Krogh
Nobel prize 1920

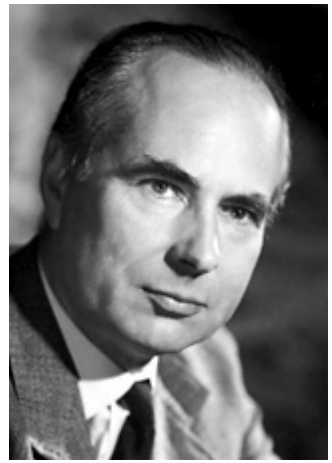
"For such a large number of problems there will be some animal of choice or a few such animals on which it can be most conveniently studied. Many years ago when my teacher, Christian Bohr, was interested in the respiratory mechanism of the lung and devised the method of studying the exchange through each lung separately, he found that a certain kind of tortoise possessed a trachea dividing into the main bronchi high up in the neck, and we used to say as a laboratory joke that this animal had been created expressly for the purposes of respiration physiology. I have no doubt that there is quite a number of animals which are similarly "created" for special physiological purposes, but I am afraid that most of them are unknown to the men for whom they were "created," and we must apply to the zoologists to find them and lay our hands on them." (Krogh, 1929)

Krogh's principle and Neuroscience research

Studying the giant axon of the squid in order to understand mechanisms of action-potential generation







Alan Hodgkin Andrew Huxley
Nobel prize 1963

Q: Why was this species chosen?

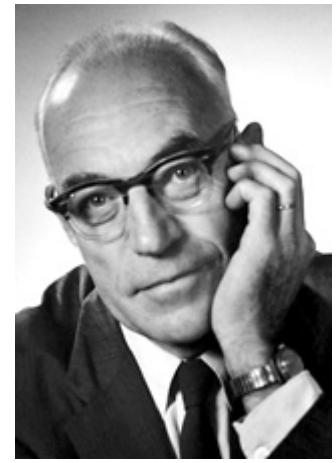
A: Because of the huge **<u>size</u>** of its axon (~1 mm diameter), which allowed using macro-wires for recording electrical potentials - and doing voltage clamp & space clamp.



Krogh's principle and Neuroscience research

Studying the frog neuromuscular junction in order to understand the physiology of synaptic transmission





Sir John Eccles
Nobel prize
1963



Sir Bernard Katz
Nobel prize
1970



Q: Why was this species chosen?

A: Because of the **size** of this synapse (end-plate) and the **simplicity** of the reflex circuit involved.

Krogh's principle and Neuroscience research

Studying the *Limulus* (horseshoe crab) retina in order to understand visual processing; discovery of the phenomenon of lateral inhibition







Haldan Hartline
Nobel prize 1967

Q: Why was this species chosen?

A: Because horseshoe crabs have long optic nerves that can be easily **split** to record from individual nerve fibers; and the retina circuitry is **simple**: the compound eye has one output neuron under each optical unit (ommatidium), facilitating the study of lateral inhibition



Krogh's principle and Neuroscience research

Studying the neurobiology of learning and memory in *Aplysia*





Eric Kandel
Nobel prize 2000



Q: Why was this species chosen?

A: Because of the **size** of its **identified** neurons; and not least importantly, because of the animal's robust **behaviors** (e.g. gill withdrawal reflex) and robust **learning** (e.g. sensitization; and classical conditioning of the gill withdrawal reflex)

Some commonly used animal models in Neuroscience:

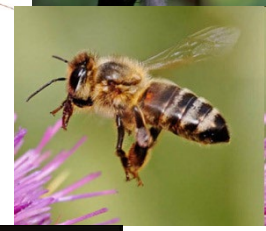
Past and Present *(Not showing less common species)*



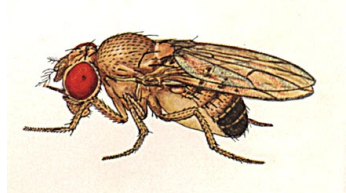






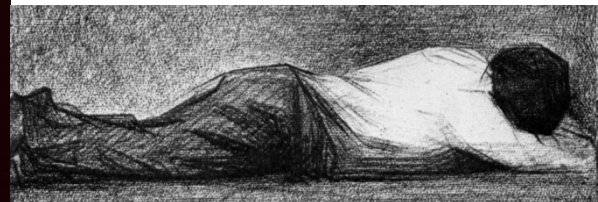








Krogh's principle vs. "standard animal models"

- **A corollary of Krogh's Principle – as viewed by Neuroethologists:** You should choose the animal species that best fits your research question (fits either in terms of the animal's *behavior* or for technical reasons) – i.e., choose well your animal model – rather than studying all the possible questions using just a few "standard animal model species" (e.g., mouse, *Drosophila*).
- **Advantages of "Standard animal models":** So much is known about their brains… Therefore, many people prefer this knowledge-base over Krogh's principle. Additionally, there are very powerful experimental tools available for these animals (particularly molecular tools).

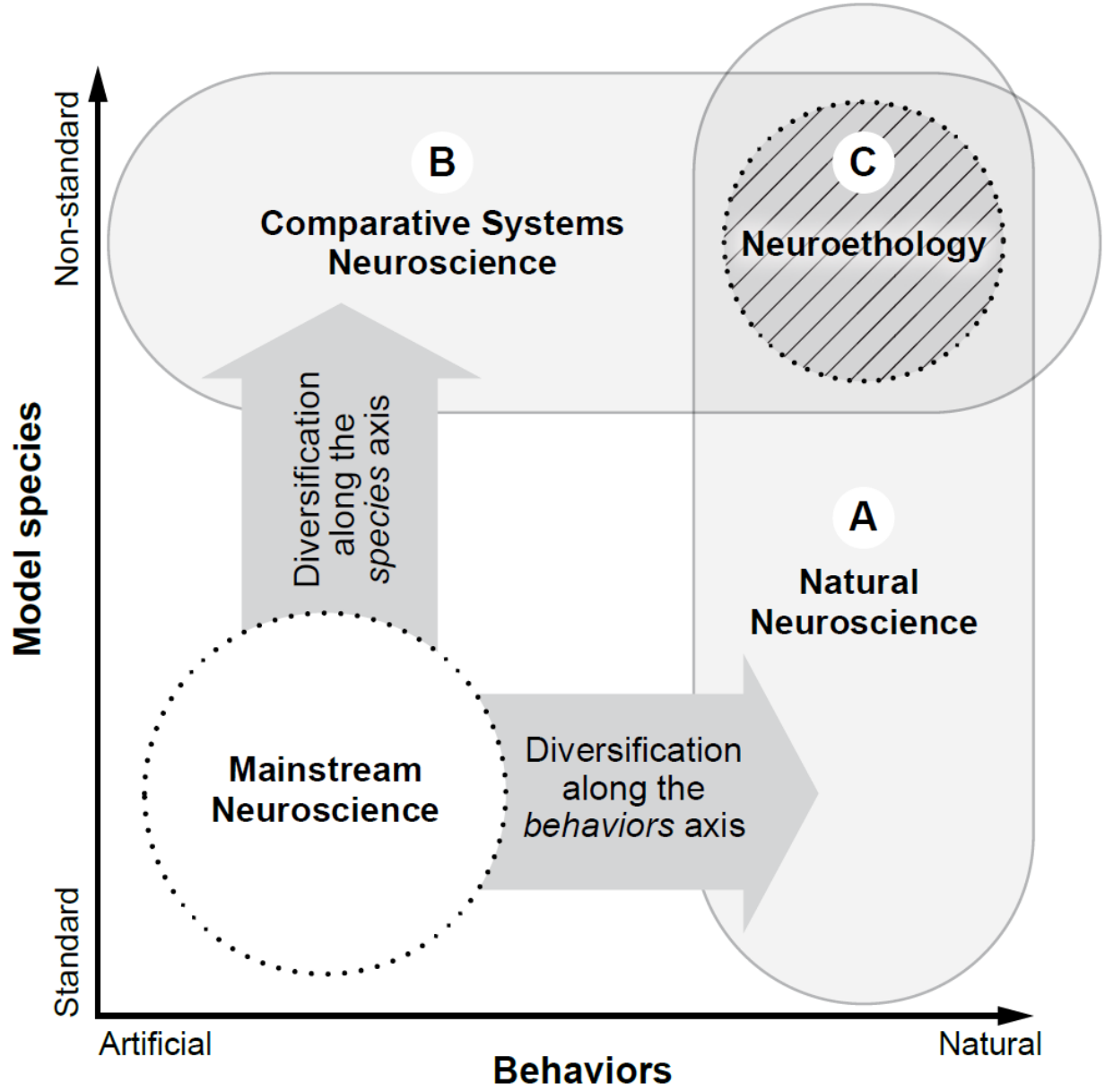
- **Discussion: Pros ? Cons ? When would you favor Krogh's principle and when favor a standard-model approach ?**

Neuroethology has some additional historical traditions, which I find more problematic

- **Focus on "champion species":**
  - Generality vs. Specificity of brain function across species
  - Relevance vs. Irrelevance to Human brain function

- **Neuroethology calls for studying the neural basis of natural behavior – but oftentimes actually studies reduced preparations** (with a few exceptions, most notably Bird Song & Neurobiology of Navigation)

Diversifying species vs. Diversifying behaviors

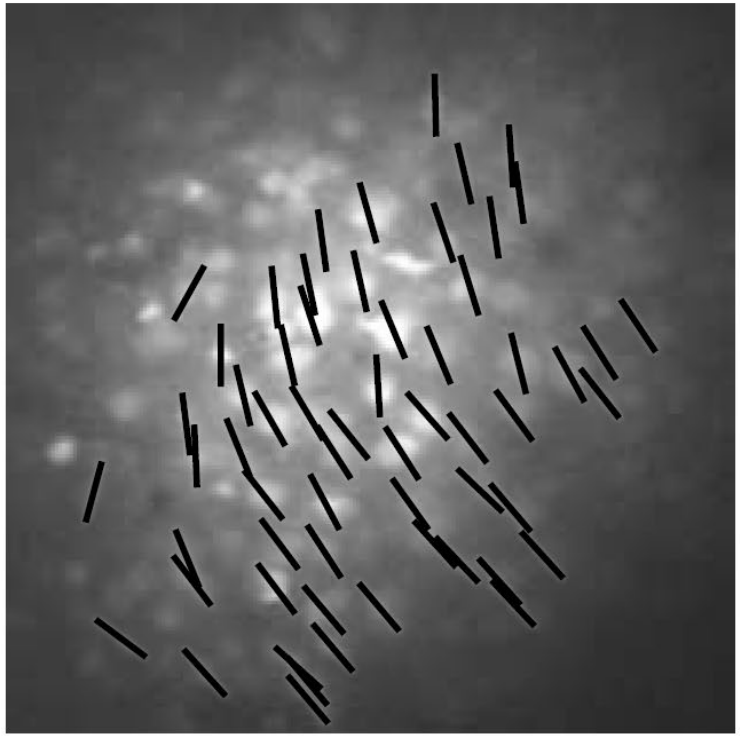


Reasons for diversifying species

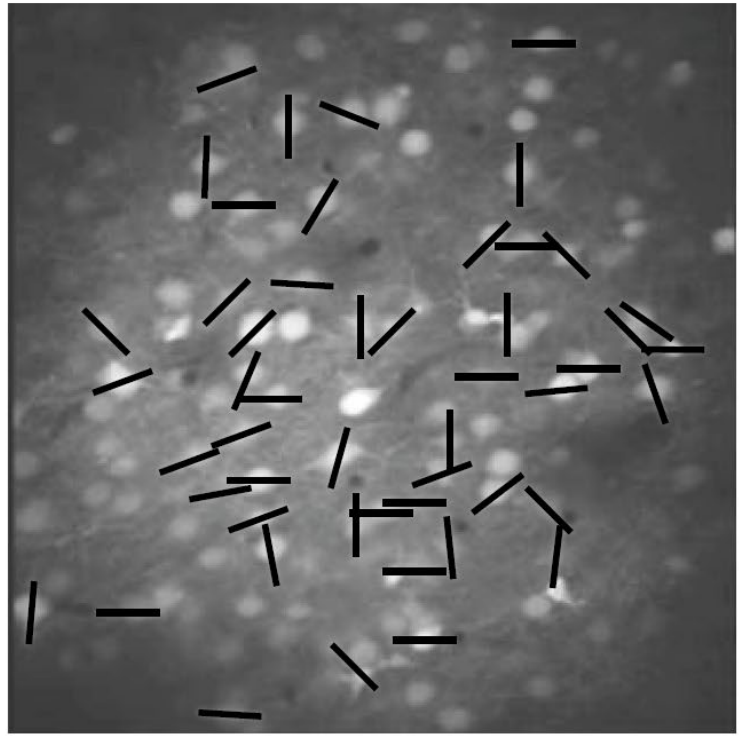
- **Comparative approach:** you can learn a lot by comparing across species

V1

Cat V1
Columnar organization



Rat V1
Salt-and-pepper organization

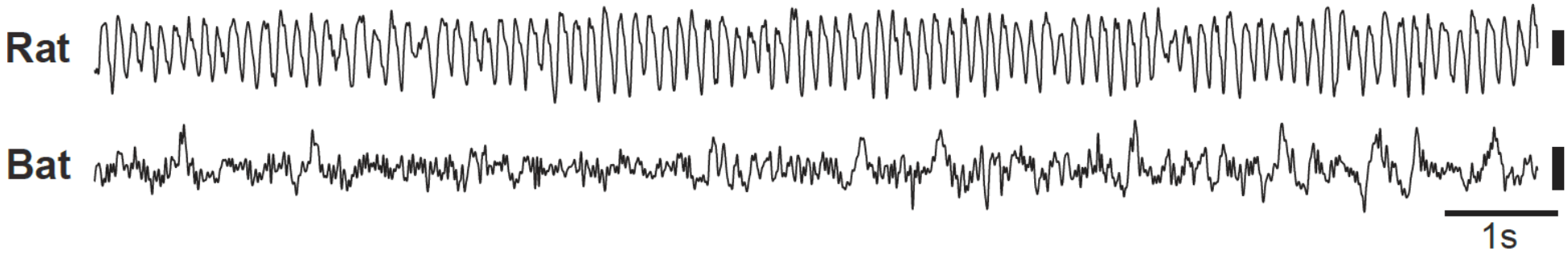


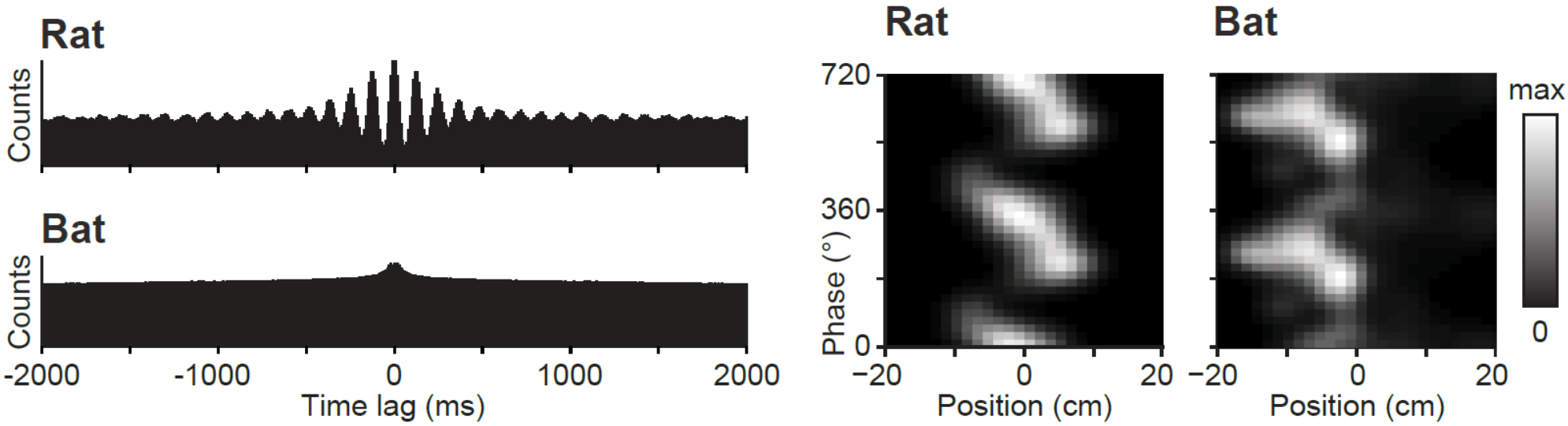
Based on Ohki et al 2005

Reasons for diversifying species

- **Comparative approach:** you can learn a lot by comparing across species

Hippocampus





Eliav et al., *Cell* (2018)

Reasons for diversifying species

- **Comparative approach:** you can learn a lot by comparing across species

Hippocampus

In rodents – 3 phenomena that are coupled together:

Rodents:	Bats
✓ Oscillations (theta)	**X** Oscillations (theta)
✓ Synchronization	✓ Synchronization
✓ Coding (of position)	✓ Coding (of position)

The comparative approach allows to identify what is invariant across species – and what is not.

* A similar result was found later in humans (Qasim et al., *Cell* 2021).

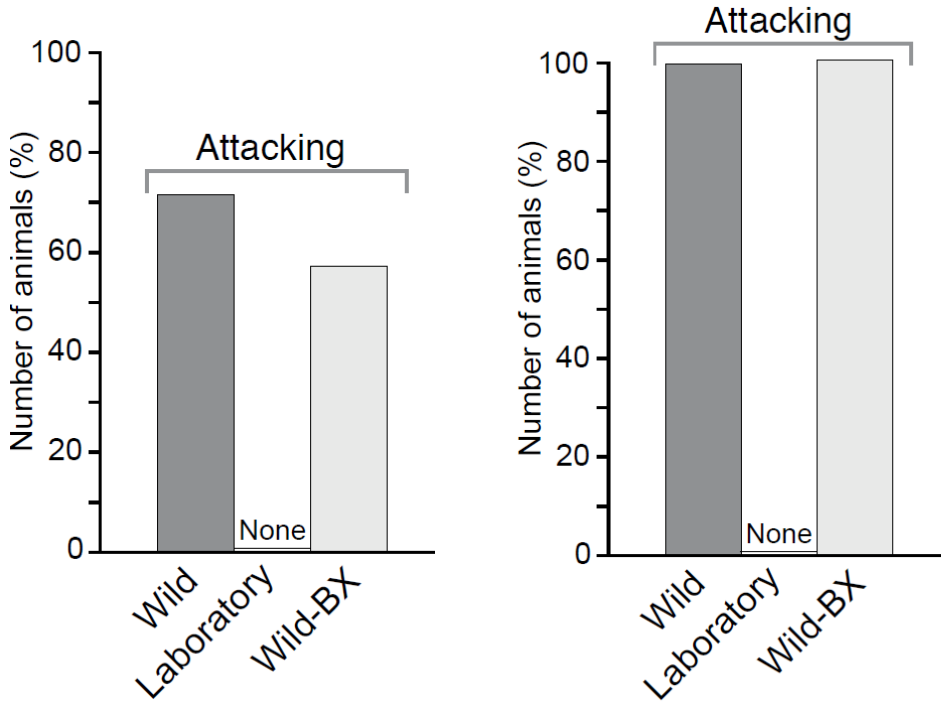
Reasons for diversifying species

- **Comparative approach:** you can learn a lot by comparing across species
  The comparative approach allows identifying mechanisms that generalize (or not) across species. Allows finding invariant computational principles (but also unique specializations).
- **Drawbacks of the pure-strain approach:** because of founder effects, mouse strains differ dramatically from each others and from wild mice

Reasons for diversifying species

- **Comparative approach:** you can learn a lot by comparing across species
- **Drawbacks of the pure-strain approach:** because of founder effects, mouse strains differ dramatically from each others and from wild mice

Dramatic differences in social behaviors



Female attacks on intruder adult females (left) and on intruder alien pups (right).

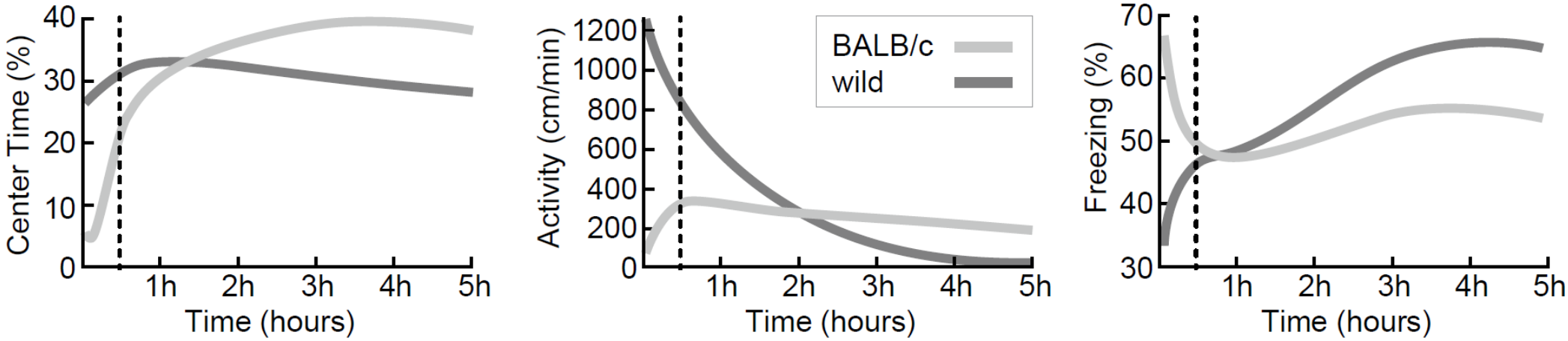
- Wild females are much more aggressive.
- **Q:** Why was aggressiveness eliminated from laboratory females during the process of mouse domestication?
- **A:** To facilitate breeding.

Based on Chalfin et al 2014 (Tali Kimchi's lab)

Reasons for diversifying species

- **Comparative approach:** you can learn a lot by comparing across species
- **Drawbacks of the pure-strain approach:** because of founder effects, mouse strains differ dramatically from each others and from wild mice

Dramatic differences in movement between BALB/c mice and wild mice



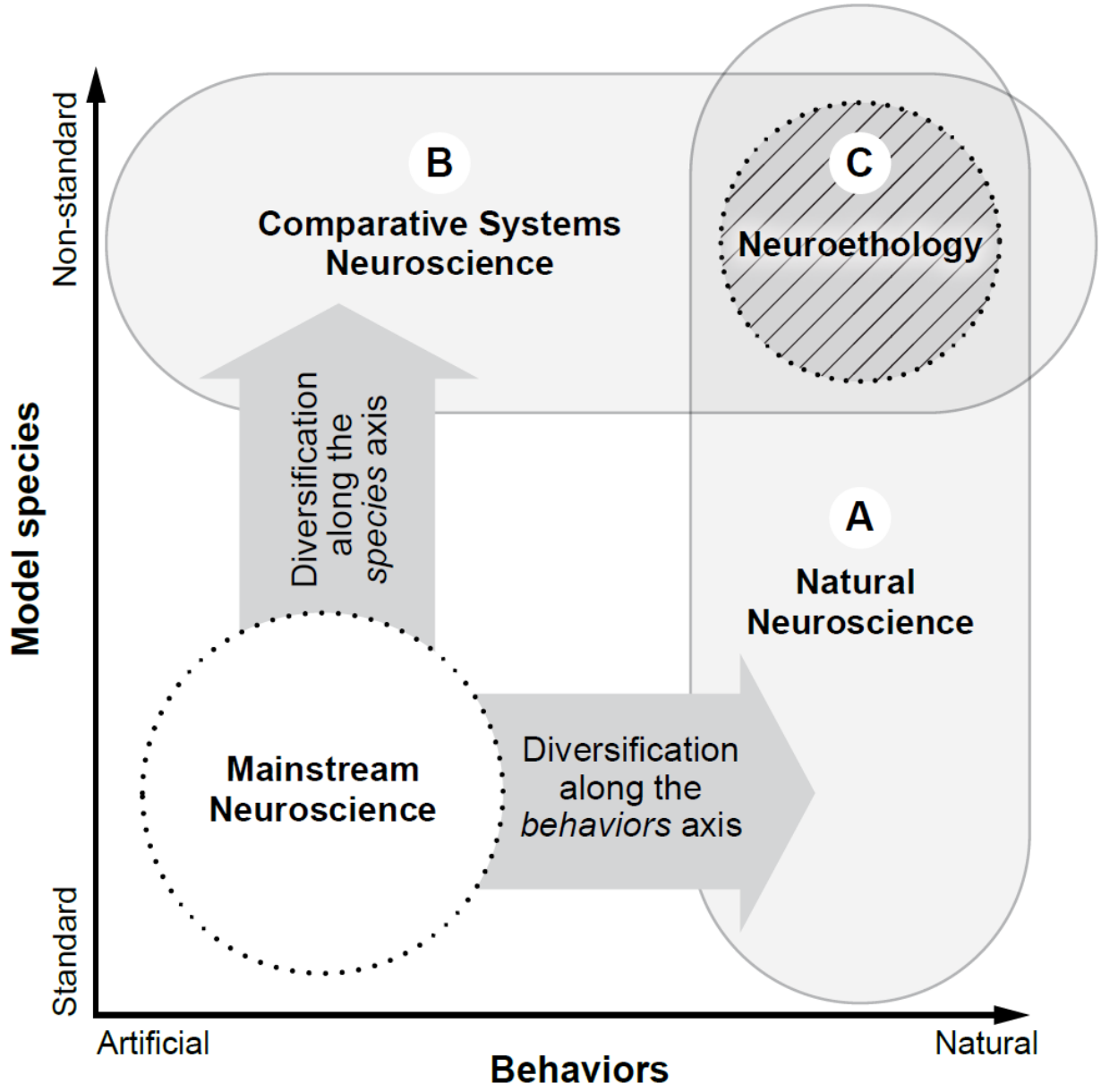
→ BALB/c mice are *not* a valid model for anxiety, as often considered !

Udi Fonio, Ilan Golani

Reasons for diversifying species

- **Comparative approach:** you can learn a lot by comparing across species
- **Drawbacks of the pure-strain approach:** because of founder effects, mouse strains differ dramatically from each others and from wild mice
- **"Sanity check": Does your finding from one species generalize to another → Also for Medical research**
- **Krogh's principle** → Different species offer unique opportunities to ask questions
- **Certain behaviors are simply not present in standard models:**
  e.g. many Social behaviors.
  Discuss: Teaching to hunt (meerkats), Altruism (bats), Theory of mind (primates), …
- **Diversity can allow identifying brain basis of specific behaviors** → e.g. Deer Mice (Hopi Hoekstra)
- **The Unknown Unknowns:** So many things were discovered in nonstandard species – who knows what may lie ahead ?

Diversifying species vs. Diversifying behaviors



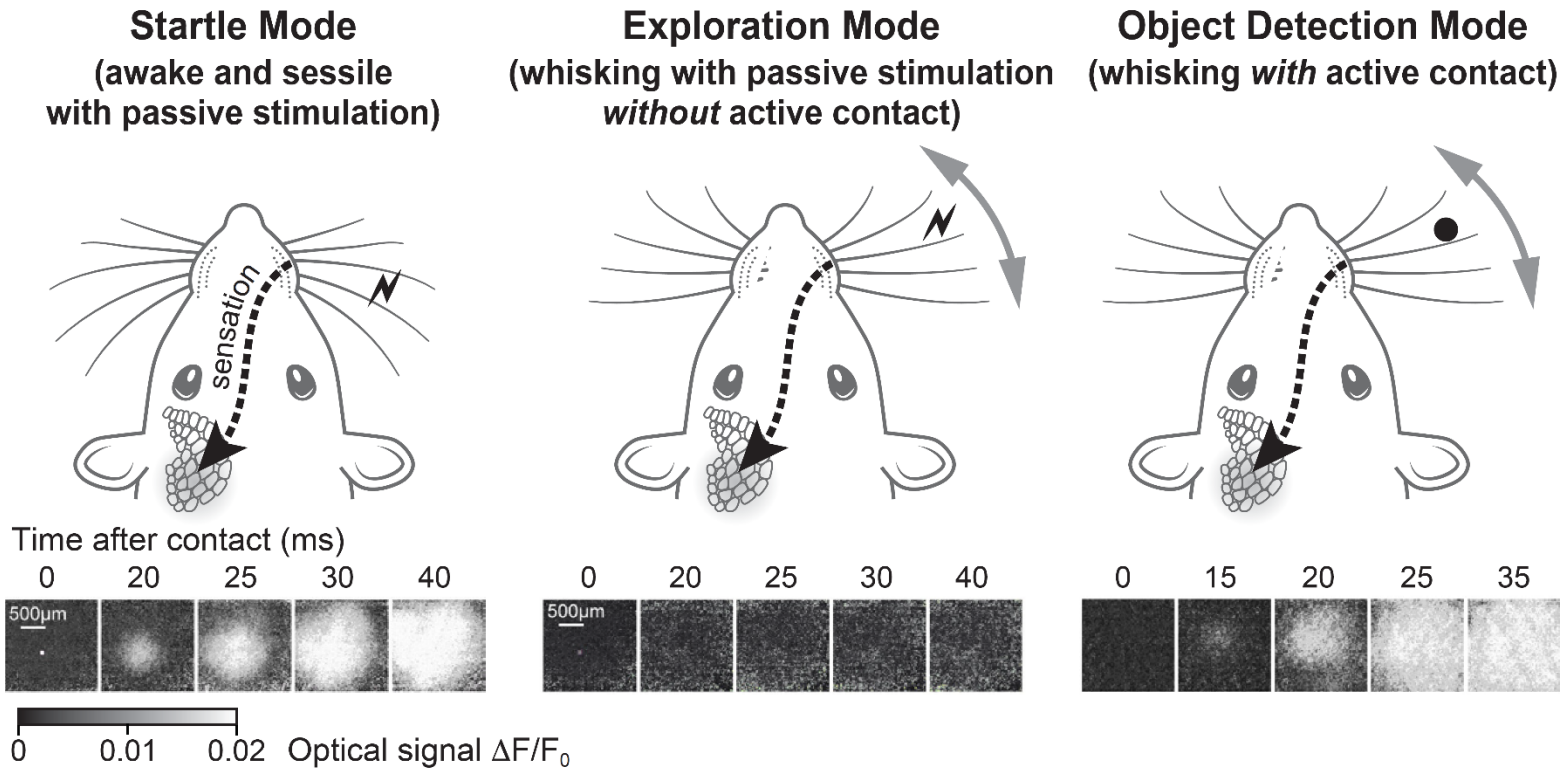
Claim:
Regardless of species used, it is important to study also natural behaviors – not just simple artificial controlled behaviors.

Reasons for diversifying to more Natural Behaviors

- **In a complex dynamical system like the brain you can't isolate and 'control' individual factors – as they are all connected**

Reasons for diversifying to more Natural Behaviors

- **In a complex dynamical system like the brain you can't isolate and 'control' individual factors – as they are all connected**
- **The brain and the world are reciprocally interacting**



Active sensing

Sensory neurons care about natural stimulation (active sensing)

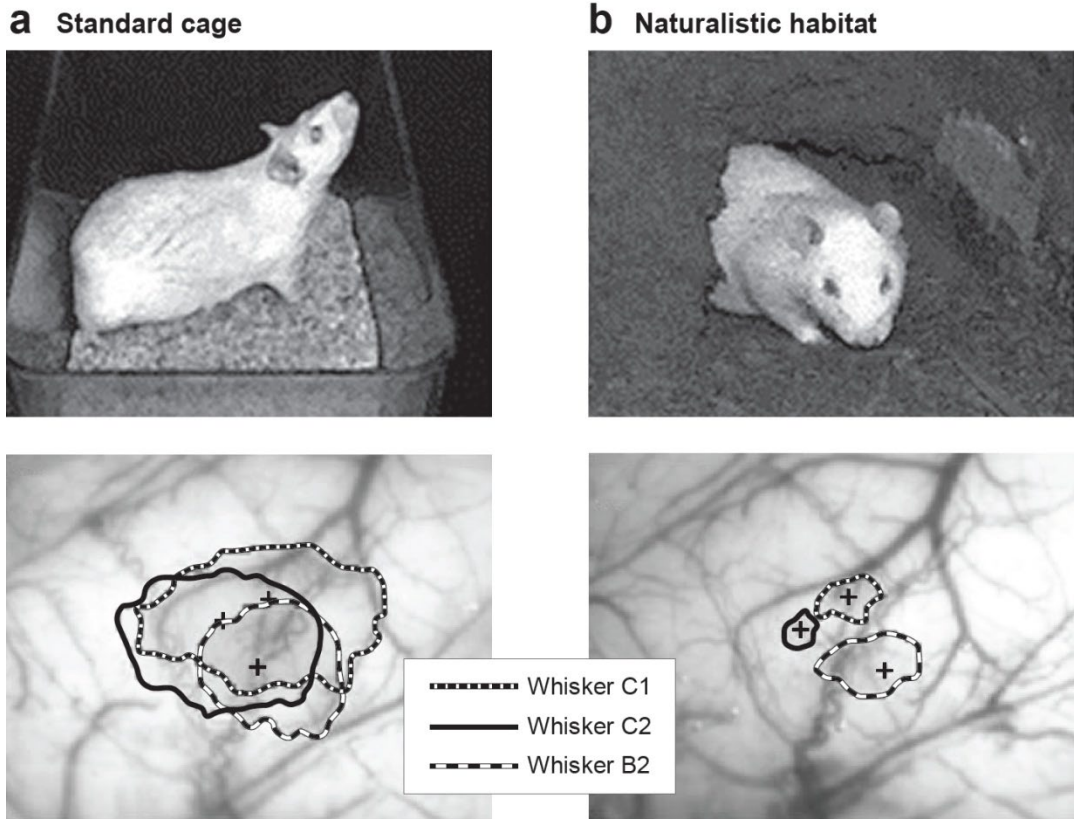
vs.

unnatural stimulation (passive)

Based on Ferezou et al. 2006

Reasons for diversifying to more Natural Behaviors

- **In a complex dynamical system like the brain you can't isolate and 'control' individual factors – as they are all connected**
- **The brain and the world are reciprocally interacting**
- **The brain responds dramatically to natural conditions**

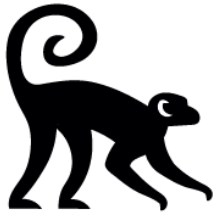


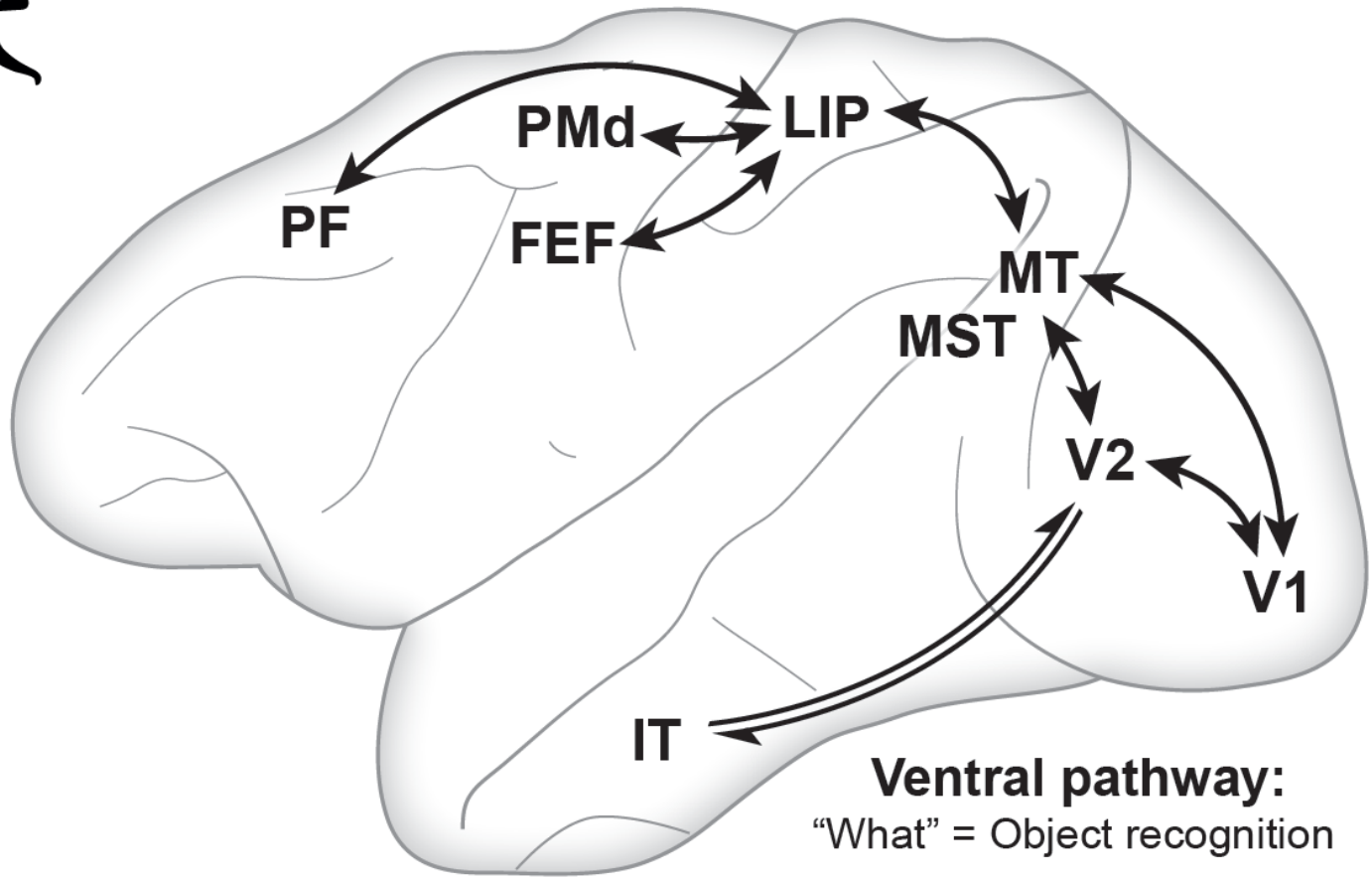
From Polley et al (2004)

Reasons for diversifying to more Natural Behaviors

- **In a complex dynamical system like the brain you can't isolate and 'control' individual factors – as they are all connected**
- **The brain and the world are reciprocally interacting**
- **The brain responds dramatically to natural conditions**
- **Many behaviors are tightly coupled to each other, and you can't isolate them naturally** (e.g. whisking and sniffing)
- **Top-down effects in the brain:** Are eliminated in many controlled experiments, but exist in natural behaviors.
  Top down anatomical projections are massive in the cortex, and many studies showed strong top-down effects of expectation, prediction, attention, etc., on neuronal activity in lower areas.

Top down projections in the cortex are usually much stronger than the bottom-up projections

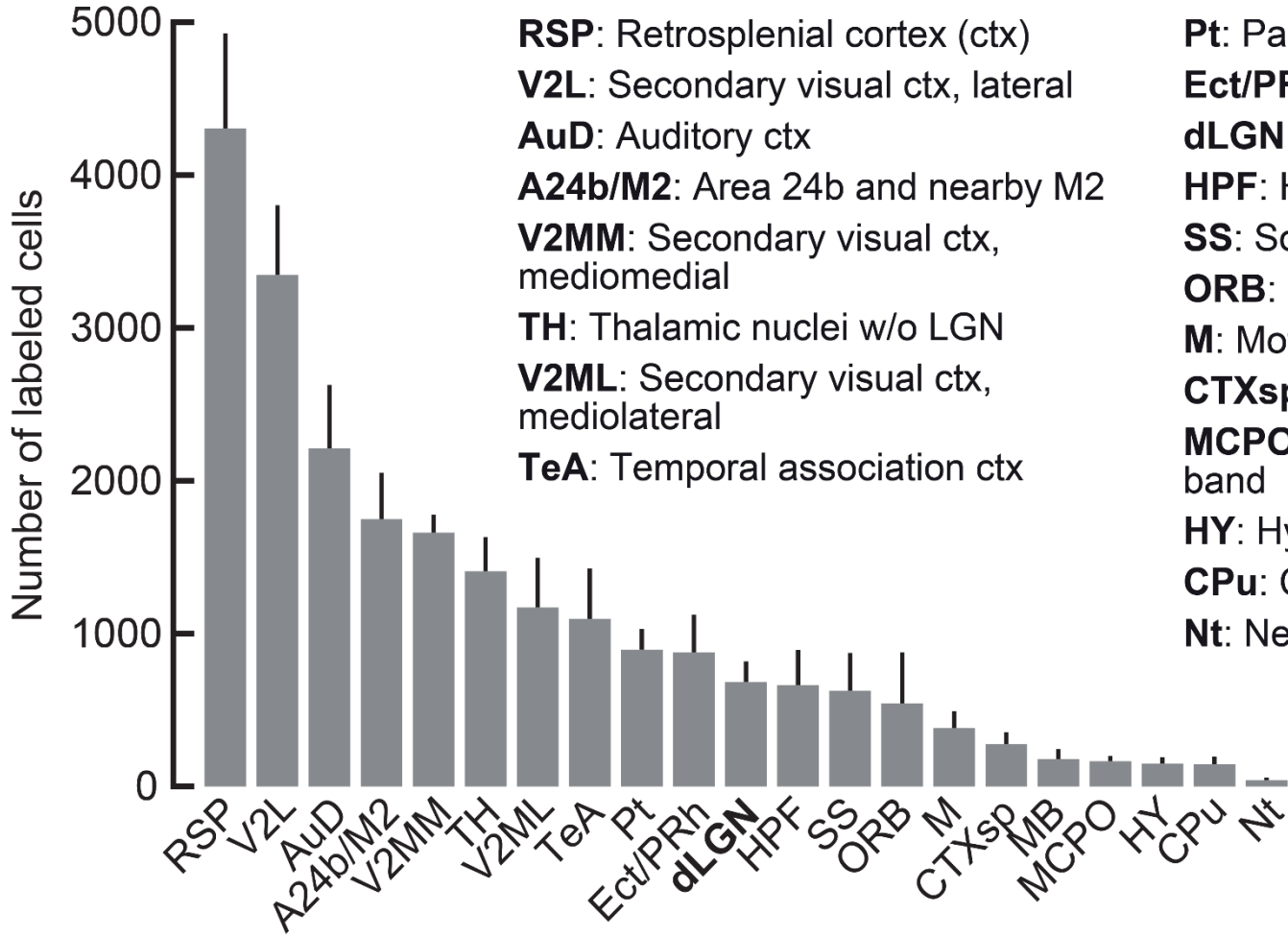




Top down projections in the cortex are usually much stronger than the bottom-up projections



V1



RSP: Retrosplenial cortex (ctx)
V2L: Secondary visual ctx, lateral
AuD: Auditory ctx
A24b/M2: Area 24b and nearby M2
V2MM: Secondary visual ctx, mediomedial
TH: Thalamic nuclei w/o LGN
V2ML: Secondary visual ctx, mediolateral
TeA: Temporal association ctx

Pt: Parietal association ctx, posterior
Ect/PRh: Ectorhinal/ Perirhinal ctx
dLGN: Lateral geniculate nucleus
HPF: Hippocampal formation
SS: Somatosensory ctx
ORB: Orbital ctx
M: Motor ctx w/o M2 and nearby A24b
CTXsp: Cortical subplate
MCPO: Magno. preoptic n. / diagonal band
HY: Hypothalamic nuclei w/o MCPO
CPu: Caudate putamen
Nt: Neural tracts

Leinweber et al 2017

Summary: Why study natural behaviors ?

- **The brain behaves very differently under natural conditions & natural behaviors:** e.g. barrel cortex, CA1 multiscale coding & CA1 social coding, responses to natural vs. artificial stimuli in sensory cortex…
- **In a complex dynamical system like the brain you can't isolate and 'control' individual factors – as they are all connected**
- **The brain and the world are reciprocally interacting**
- **Top-down effects in the brain** (expectation, attention, emotions…) are eliminated in many controlled experiments, but exist in natural behaviors
- **Reducing stress**: Some laboratory tasks are stressful for the animal. Besides ethical concerns, stress affects brain function & structure
- **Reducing overtraining:** Overtraining affects brain function & structure
- **Evolution:** The brain has evolved to do specific behaviors – and therefore these natural behaviors should be our focus of research
- **Relevance:** We need to study behaviors that are relevant for the animal
- **Theoretical work suggests it's essential to use high-dimensional behaviors** to assess the true dimensionality of neuronal dynamics
- **The problem of Unknown Unknowns**

Neural Basis of Behavior … But which Behavior(s) ?

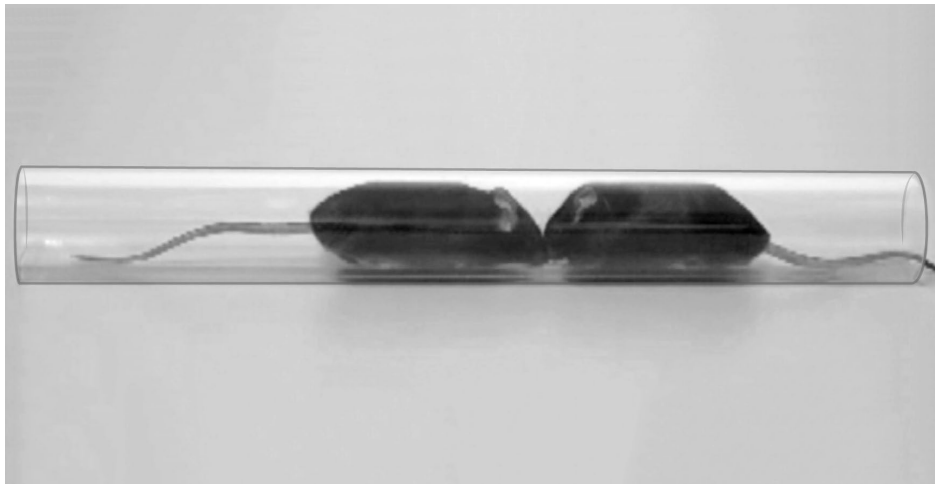
Controlled laboratory experiments



Natural behaviors

"Controlled but Unnatural"

"Natural but uncontrolled"

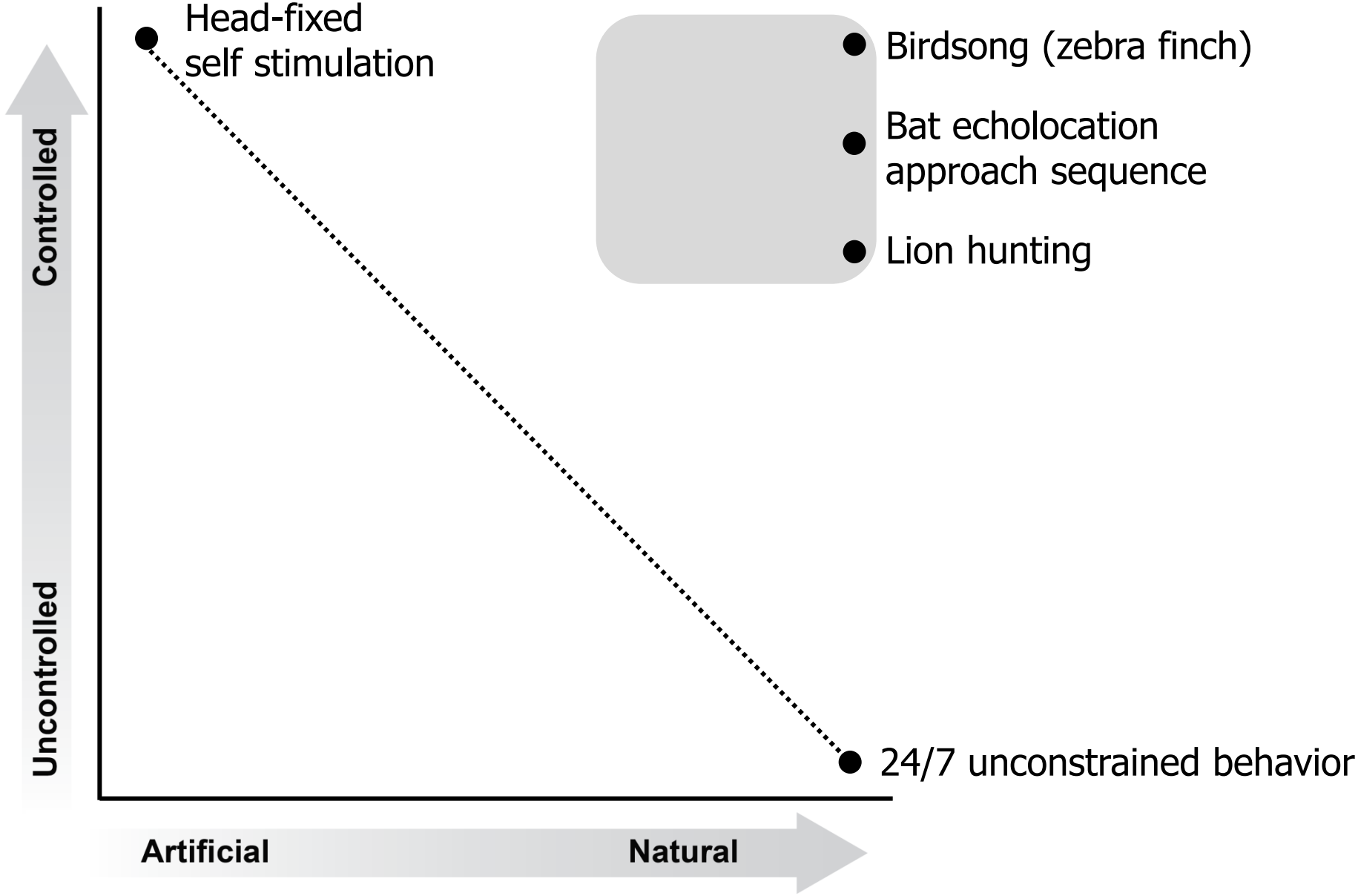


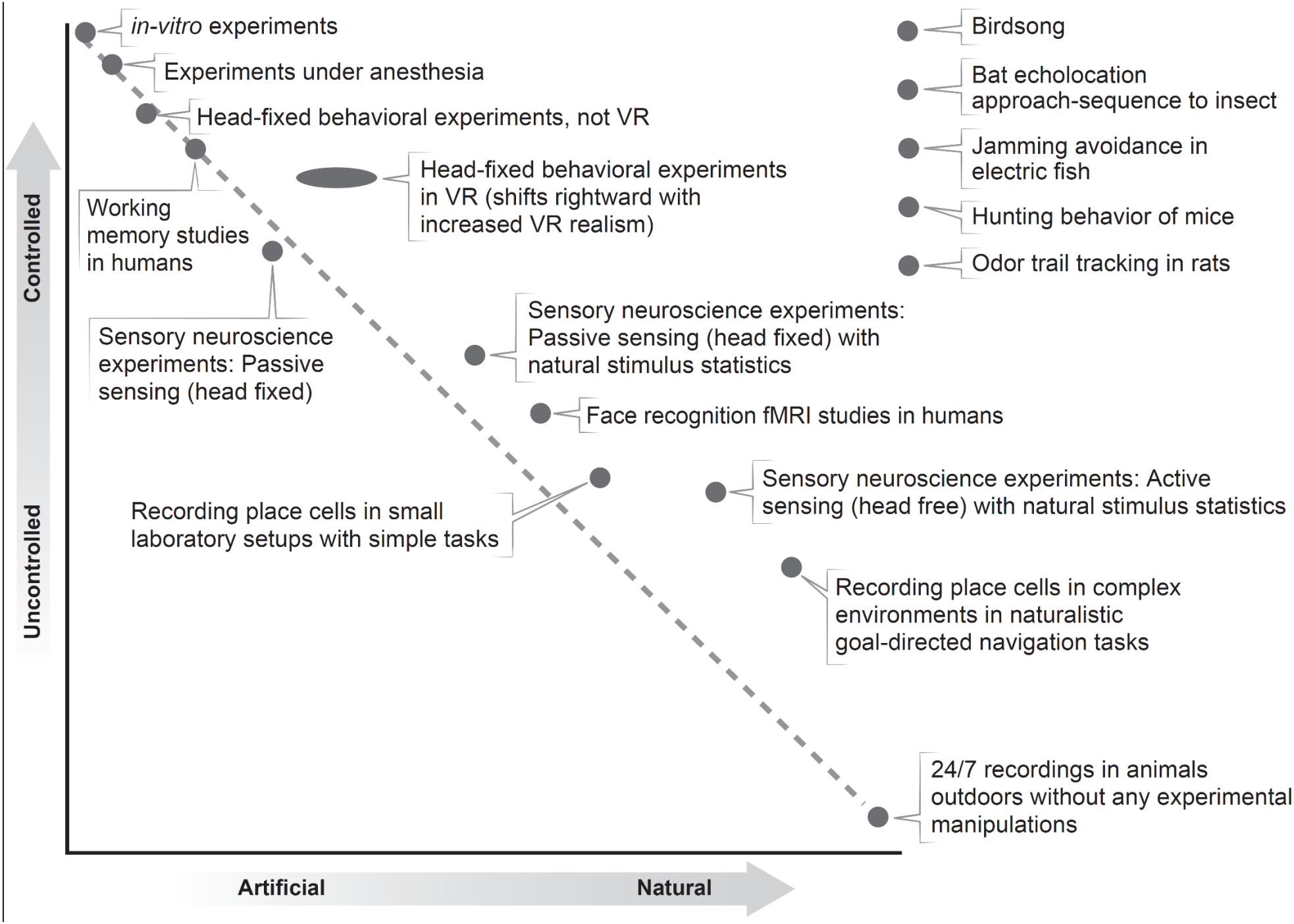
Tube test in mice



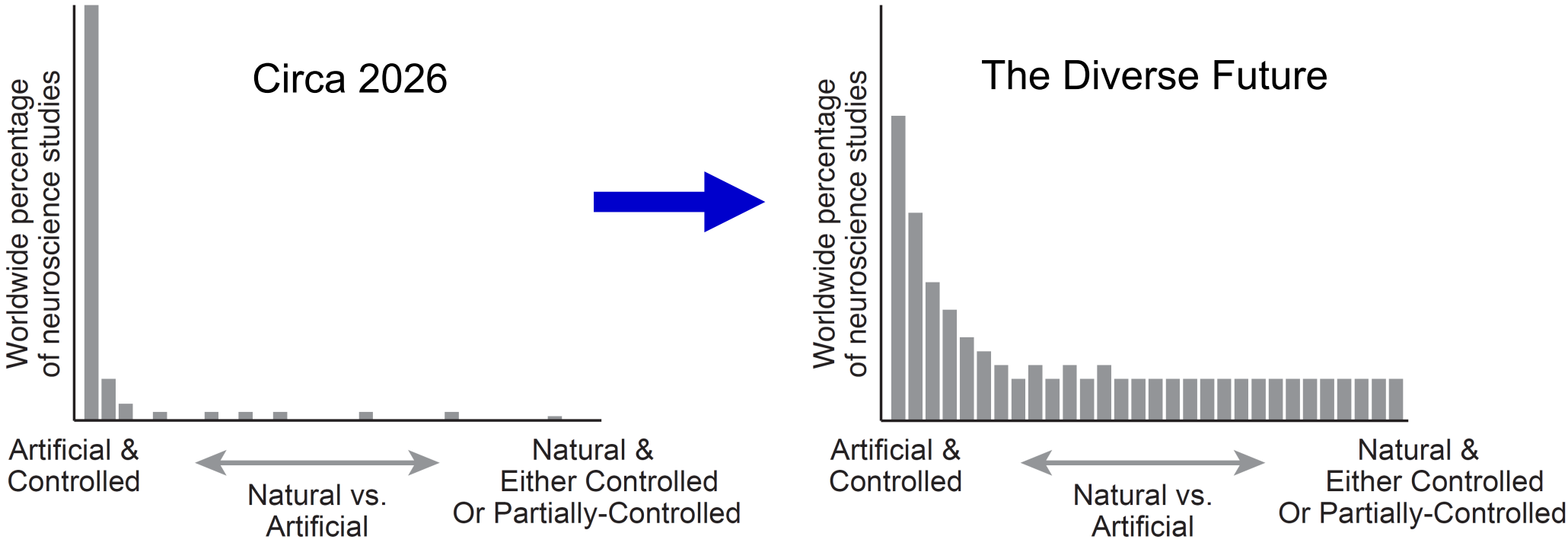
Group hunting in lions &
Communal defense in buffalo

Neural Basis of Behavior … But which Behavior(s) ?



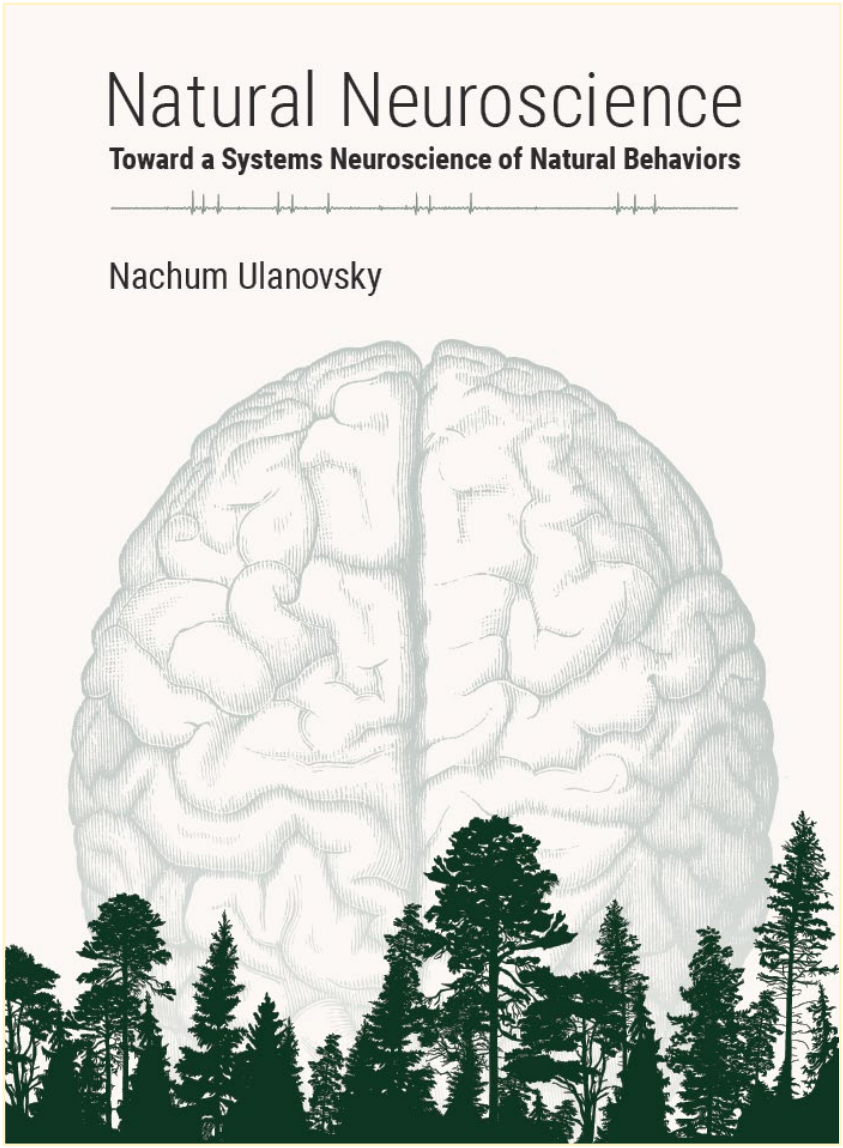

in-vitro experiments
Experiments under anesthesia
Head-fixed behavioral experiments, not VR
Working memory studies in humans
Head-fixed behavioral experiments in VR (shifts rightward with increased VR realism)
Sensory neuroscience experiments: Passive sensing (head fixed)
Sensory neuroscience experiments: Passive sensing (head fixed) with natural stimulus statistics
Face recognition fMRI studies in humans
Recording place cells in small laboratory setups with simple tasks
Sensory neuroscience experiments: Active sensing (head free) with natural stimulus statistics
Recording place cells in complex environments in naturalistic goal-directed navigation tasks
24/7 recordings in animals outdoors without any experimental manipulations
Birdsong
Bat echolocation approach-sequence to insect
Jamming avoidance in electric fish
Hunting behavior of mice
Odor trail tracking in rats
Controlled
Uncontrolled
Artificial
Natural

Schematic vision of the future…



Take Home Message

- **Natural Behaviors vs. Reductionism: ↑ Diversity needed.**



Topics:

- Active sensing
- Sensory neuroscience - Natural stimuli
- Neurobiology of navigation
- Real-life Memory research
- Decision making - in the real world
- Social neuroscience
- Technological advances enabling a Natural Neuroscience approach
- Synthesis of Behavioral Neuroscience, Behavioral Ecology & Ecological Psychology

Ecological Validity –
Not necessarily Outdoors !

MIT Press – 2025